CDEQ V5.0: A Universal Framework for Critical Transition and Global Self-Sustained Dynamics 临界相变与全局自持动力学通用框架

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Abstract

CDEQ V2.0 provides convergence, braking and cage mechanism for global stability.
CDEQ V3.5 defines critical transition and self-sustained dynamics as universal laws.
CDEQ V5.0 is the final unified version: V2.0 Brake + V3.5 Throttle + GCVT-10 Mandatory Verification.

This paper presents the only legally valid, globally unified, plain-text, non-conflicting mathematical system of CDEQ V5.0. All formulas, symbols and structures are fixed and unchangeable. The framework applies to all dynamical domains and fully obeys the 16 classical axioms of physics and thermodynamics.

Keywords:
CDEQ V5.0, PCTT-CW, critical transition, self-sustained dynamics, dynamical equilibrium, GCVT-10, topological stability, brake module, throttle module, safety interlock

1. Introduction

1.1 Version Definition (Unchangeable)

- V2.0 = Convergence + Brake + Cage (stability only)
- V5.0 = V2.0 Brake + V3.5 Throttle + GCVT-10 (full controllable dynamics)

1.2 Foundational Topology

All CDEQ systems are built on PCTT-CW (Penta-Cyclic Clockwise Topology), the minimal and optimal topology for distributed non-antagonistic stability.

2. Axiom System (16 Axioms, Full Inheritance)

2.1 Thirteen Universal Axioms

1. Energy Conservation (First Law of Thermodynamics)
2. Entropy Increase Principle (Second Law of Thermodynamics)
3. Third Law of Thermodynamics
4. Zeroth Law of Thermodynamics
5. Newton’s First Law (Inertia)
6. Newton’s Second Law (Force–Motion Relation)

7. Newton's Third Law (Action–Reaction)
8. Conservation of Momentum
9. Conservation of Mass
10. Ideal Gas Law (PV = nRT)
11. Lyapunov Stability Axioms
12. Feedback Control Theory Principles
13. Mathematical Convergence and Limit Axioms

2.2 Three Boundary Axioms

14. Initial external energy input is required (no perpetual motion).
15. Self-sustained power comes only from internal or environmental energy recovery.
16. Mode transition requires GCVT verification for 10 consecutive steps.

All 16 axioms are preserved without any change.

3. CDEQ V5.0 Core Mathematical System

3.1 V2.0 Basic Brake Formula

$\Psi' = \text{clip}[\ \varphi\Psi_1 - \varphi\Psi_2 + \omega\xi\ ]$
$\rightarrow \Phi<\Gamma\ \&\ |E_{11}-5|<\varepsilon\ \&\ |\Sigma-15|<\varepsilon\ \forall 10$

3.2 【Only Legally Valid · Globally Unified · Unchangeable】

V5.0 Ultimate Harpoon Formula

$\Psi' = \text{clip}[\ \varphi\Psi_1 - \varphi\Psi_2 + \omega\xi\ ] + (\Psi\geq\Psi\pi\ \&\ \Pi\text{self}\geq\Pi\text{ext})\cdot\nabla C$
$\rightarrow \Phi<\Gamma\ \&\ |E_{11}-5|<\varepsilon\ \&\ |\Sigma-15|<\varepsilon\ \forall 10$

3.3 Three Core Modules (Original Symbols Only)

1. $\text{clip}[\ \varphi\Psi_1 - \varphi\Psi_2 + \omega\xi\ ]$
= V2.0 Brake Module (convergence, cage, stability)
2. $(\Psi\geq\Psi\pi\ \&\ \Pi\text{self}\geq\Pi\text{ext})$
= V3.5 Throttle Switch
3. ∇C
= V3.5 Throttle Drive (thrust, acceleration, self-sustainment)
4. $\Phi<\Gamma\ \&\ |E_{11}-5|<\varepsilon\ \&\ |\Sigma-15|<\varepsilon\ \forall 10$
= GCVT 10-step mandatory verification

4. V5.0 Segmented State Machine (Plain Text Only)

$\Psi' = 0$,
Condition: $\Psi<\Psi\pi\ \&\ \neg\text{GCVT}$
[Unqualified · Fully Closed]

$\Psi' = \text{clip}[\varphi\Psi_1 - \varphi\Psi_2 + \omega\xi] + \nabla C_{sub}$,
Condition: $\Psi \geq \Psi\pi$ & GCVT & $\Pi_{self} < \Pi_{ext}$
[V2.0 Brake + Subcritical Throttle]

$\Psi' = \text{clip}[\varphi\Psi_1 - \varphi\Psi_2 + \omega\xi] + \nabla C_{crit}$,
Condition: $\Psi \geq \Psi\pi$ & GCVT & $\Pi_{self} \geq \Pi_{ext}$ & $\neg$SS
[V2.0 Brake + Critical Throttle]

$\Psi' = \text{clip}[\varphi\Psi_1 - \varphi\Psi_2 + \omega\xi] + \nabla C_{sust}$,
Condition: $\Psi \geq \Psi\pi$ & GCVT & $\Pi_{self} \geq \Pi_{ext}$ & SS
[V2.0 Brake + Fully Self-Sustained Throttle]

Unified Constraint for All Modes:
$\Phi < \Gamma$ & $|E_{11}-5| < \varepsilon$ & $|\Sigma-15| < \varepsilon$ $\forall 10$

5. GCVT-10: The Global Safety Interlock

GCVT-10 is NOT a brake, nor a throttle.
It is the mandatory safety interlock that prevents unsafe activation.

- The interlock is locked by default → GCVT-10 is not passed by default.
- The interlock must be fully released → GCVT requires 10 consecutive valid steps.
- The drive system activates only after interlock release → ∇C only activates after GCVT-10 passes.
- Without interlock release, drive command is ignored → Without GCVT-10, throttle ∇C is forced to zero.

A system without safety interlock cannot be safely deployed.
A self-sustained system without GCVT-10 cannot be certified for engineering use.

GCVT-10 eliminates:

- False activation from transient stability
- Unintended operation from noise or disturbance
- Unreliable self-sustainment from incomplete convergence

Only after 10 consecutive valid steps can the system be confirmed as truly stable, safe, and ready for self-sustained operation.

This is the core design philosophy of CDEQ V5.0:
Brake holds the system stable, throttle provides controlled drive, GCVT grants final safety permission.

6. Universal Domain Applications

CDEQ V5.0 is domain-invariant. Only ∇C changes:

- Aerospace: Ψ = velocity
- Materials: Ψ = coverage
- Chemistry: Ψ = temperature
- Learning: Ψ = practice count
- Economy: Ψ = capital
- Robotics: Ψ = energy harvest rate

7. Version Hierarchy

1. PCTT-CW: Eternal topological crust
2. CDEQ V2.0: Brake / Cage core
3. CDEQ V3.5: Throttle / Self-sustainment core
4. CDEQ V5.0: Brake + Throttle + GCVT-10 (final unified framework)
5. Future versions: domain adaptations only

8. Conclusion

CDEQ V5.0 is the final unified framework for critical transition and global self-sustained dynamics.
It integrates:

- V2.0 Brake (always online)
- V3.5 Throttle (conditionally activated)
- GCVT-10 (mandatory safety interlock)

All formulas, symbols and structures are globally unique, legally valid and unchangeable.

Global self-sustainment, controllable convergence, eternal topology.

9. Engineering Implementation

9.1 FPGA/ASIC Realization of CDEQ V5.0

This is a minimal functional demonstrator for CDEQ V5.0, implementing the core PCTT-CW topology, V2.0 brake, V3.5 throttle, and GCVT-style safety interlock.
This design is for functional verification only. No precision guarantee is provided.

verilog
```
module cdeq_v5_paper(
    input clk,
    input rst_n,
    output reg [15:0] p0,p1,p2,p3,p4,
    output reg converged
);

localparam Q14 = 16384;
localparam COEF_S = 9800;
```

```
localparam COEF_C = 6500;
localparam NOISE_GAIN = 2000;
localparam THRESH = 10000;
localparam DRIVE = 1000;
localparam STEADY = 4;
localparam INIT = 8192;

localparam [2:0] PREV [0:4] = '{4,0,1,2,3};
localparam [2:0] PREV2[0:4] = '{3,4,0,1,2};

reg [15:0] p_hist[0:1][0:4];
reg [15:0] p_next[0:4];
reg [3:0] cnt;

reg [31:0] lcg;
wire [15:0] xi;

always @(posedge clk or negedge rst_n) begin
	if(!rst_n)
		lcg <= 32'd123456789;
	else
		lcg <= (lcg * 32'd1103515245 + 32'd12345) & 32'h7FFFFFFF;
end

assign xi = (lcg % 4096) - 2048;

always @(*) begin
	integer i;
	reg [15:0] avg;
	reg en;

	avg = (p_hist[1][0]+p_hist[1][1]+p_hist[1][2]+p_hist[1][3]+p_hist[1][4])/5;
	en = (avg >= THRESH) && (cnt >= STEADY);

	for(i=0; i<5; i=i+1) begin
		reg [31:0] t1,t2,t3,con;
		t1 = COEF_S * p_hist[1][PREV[i]];
		t2 = COEF_C * p_hist[1][PREV2[i]];
		t3 = NOISE_GAIN * xi;
		con = (t1 - t2 + t3) >> 14;
		con = con < 0 ? 0 : con > Q14 ? Q14 : con;
		p_next[i] = con + (en ? DRIVE : 0);
	end
end

always @(posedge clk or negedge rst_n) begin
```

```
    if(!rst_n) begin
        p_hist[0] <= '{INIT,INIT,INIT,INIT,INIT};
        p_hist[1] <= '{INIT,INIT,INIT,INIT,INIT};
        cnt <= 0;
        converged <= 0;
    end else begin
        p_hist[0] <= p_hist[1];
        p_hist[1] <= p_next;
        cnt <= cnt < STEADY ? cnt + 1 : cnt;
        converged <= cnt >= STEADY;
    end
end

always @(*) begin
    p0 = p_hist[1][0];
    p1 = p_hist[1][1];
    p2 = p_hist[1][2];
    p3 = p_hist[1][3];
    p4 = p_hist[1][4];
end

endmodule
```

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